

CHENGLONG YU

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EDUCATION

Institute of Computing Technology, Chinese Academy of Sciences (ICT, CAS) Beijing, China
Master of Engineering in Computer Technology Sep. 2024 – Expected Jun. 2027
China University of Petroleum (East China) Qingdao, China
Bachelor of Engineering in Computer Science and Technology Sep. 2020 – Jun. 2024
GPA: 4.12/5 | Rank: 3/114 (Top 3%)

RESEARCH INTEREST

Reinforcement Learning, Knowledge Distillation, Diffusion Models, World Models, Resource Allocation

RESEARCH EXPERIENCE

Joint Task Offloading and Resource Allocation with Multi-Task Energy Control based on World Model-Guided Collaborative Multi-Agent Deep Reinforcement Learning

Independent Research Dec. 2023 - Nov. 2025

- **Proposed** a multi-task energy control constrained joint optimization scheme for the task offloading and resource allocation (MEC-JTORA) to maximize the terminal energy efficiency.
- **Designed** an error-corrected energy consumption model (ECECM) to allocate appropriate energy to tasks.
- **Developed** a World Model-guided Collaborative Multi-Agent Deep Reinforcement Learning (WM-MADRL) algorithm, leveraging the world model to accelerate convergence and guide policy exploration.
- **Achieved** a 36%~89% improvement in convergence speed over baseline algorithms, and increased the total size of completed tasks and energy efficiency by 4~9 times compared to systems with equal energy allocation.

Sample-Efficient Data-Free Knowledge Distillation

Independent Research Aug. 2025 - Present

- **Proposed** a teacher-feedback-driven mechanism to dynamically optimize static GAN embeddings, aligning text priors with the original data distribution.
- **Designed** a batch feature repulsion strategy for intra-class diversity and a sub-optimal class elevation mechanism for hard sample generation, improving synthesis efficiency.
- **Combined** offline pre-trained diffusion models with online GANs for data augmentation, enriching image semantics while bypassing high diffusion inference latency.
- **Achieved** competitive or even surpassing SOTA accuracies on CIFAR-10/100 by distilling student models via the proposed framework.

PUBLICATIONS & PATENTS

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- **C.Yu**, W. Xing, J. Shi, Y. Zhou and L. Liu, "Collaborative Multi-Agent Deep Reinforcement Learning for Joint Task Offloading and Resource Allocation with Long Term Energy Control," 2025 IEEE 102nd Vehicular Technology Conference (VTC2025-Fall), Chengdu, China, 2025, pp. 1-5. [pdf]
 - **C.Yu**, Y. Zhou, Y. Qi and J. Wang, "Joint Task Offloading and Resource Allocation with Multi-Task Energy Control based on World Model Guided Collaborative Multi-Agent Deep Reinforcement Learning," IEEE Internet of Things Journal (under review).
 - Y. Zhou, **C. Yu**, J. Shi, Y. Qi, W. Xing, and L. Liu, "An Industrial Internet System for Resource Allocation Based on Collaborative Multi-Agents." Chinese National Invention Patent, Patent No.: CN202511002125.2.

HONORS & AWARDS

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- Outstanding Graduate of Shandong Province - 2024
 - National Encouragement Scholarship (Three times) - 2021, 2022, 2023
 - UCAS Academic Scholarship (Twice) - 2024, 2025
 - Honors Bachelor's Degree, China University of Petroleum (East China) - 2024

TECHNICAL SKILLS

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- **Technology stack:** Python, C, C++, Pytorch, Linux, Git, CUDA